AI Candidate Ranking System

E2E Pitch Deck & Evaluation Report

Problem Statement

- Traditional ATS filters rely on simple keyword matching and are easily gamed.
- Candidates engage in keyword-stuffing (e.g. listing Python/AI skills with no experience).
- Mismatches exist between candidate titles and actual career description narratives.
- Scalability issues: processing 100,000+ profiles manually or with heavy LLMs is impossible.

System Architecture

1. Data Ingestion: Stream JSONL to handle large 487MB dataset in constant memory.
2. Fast Semantic Filter: Compute lightweight Jaccard/Sentence Embeddings on all candidates.
3. Deep Scorer: Calculate precise skill fit weights, verification penalties & behavioral signals.
4. LLM Evaluator: Batch calls to DeepSeek API (with rate limiters) for narrative reasonings.
5. Fallback Mode: Seamless transition to rule-based templates if DeepSeek is offline.

Scoring Methodology

Final Score = 0.40 * Technical + 0.35 * Career + 0.25 * Behavioral

• Technical (40%): Semantic similarity, required skill overlap, core tools.

• Career (35%): Experience (target 5-9, peak 6-8), current title, product company history.

• Behavioral (25%): Login recency, open to work status, response rate, notice period.

• Hard Filters & Honeypots: Auto 0.0 score (consulting-only, academic, impossible YoE).

Key Data Insights

- Total candidates processed: 100
- Average candidate composite score: 0.4561
- Highly endorsed skills: Python, PyTorch, SQL.
- Notice period penalty successfully applied to long notice profiles.

Top 5 Ranked Candidates

Candidate ID	Composite Score	Reasoning Summary
CAND_0008295	0.5846	A dedicated AI Research Engineer with a 6.5-year track record....
CAND_0054859	0.5805	A dedicated Senior Software Engineer (ML) with a 6.9-year trac...
CAND_0010149	0.5676	A dedicated ML Engineer with a 6.9-year track record. They exc...
CAND_0088025	0.5646	A dedicated Staff Machine Learning Engineer with a 8.6-year tr...
CAND_0080766	0.5475	Evaluating this Staff Machine Learning Engineer profile reveal...